Jianlun (Lody) Sun, MSc.

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Research Interest

I am interested in how the brain constructs and maintains meaning over time. Experience arrives continuously, yet we perceive it as structured: segmented into events, linked by causal and temporal relations, and organized into wholes that can be recalled, revised and told to others. My work asks how these structures are built during ongoing experience, how they are represented in the brain, and how they are transformed as perception becomes memory. I approach this with naturalistic paradigms and computational methods, combining behavioral modeling, EEG, fMRI and language-model analysis, and I take up new methods when a question requires them.

Education

Sep 2024 – Jul 2026 **M.Sc. Neuroscience, University of Helsinki** (GPA: 4.28/5)
Thesis title: *Staged Rule Acquisition in WishGLD: Dissociation and Compression Across Learning*. (Grade: 4/5)

- Thesis: trial-resolved analysis of a naturalistic set-shifting task (WishGLD), combining behavioral modeling, two-state HMM trajectory markers and feedback-locked EEG. Showed that rule acquisition is not a single event: hypothesis collapse, neural confirmation at the first correct response and behavioral stabilization can dissociate, and this staged structure compresses with task exposure.

Sep 2018 – Jun 2022 **B.Sc. Biological Sciences, ShanghaiTech University**
Thesis title: *Exploring the relationship between Locus Coeruleus integrity and set-shifting dysfunction in Parkinson's Disease*.

Research Publications

Conference & Symposium Presentations

1. J. Sun, "When payoff returns: Reinstatement of earlier narrative context during recall," in *Brain & Mind Symposium 2026*, Poster presentation, Helsinki, Finland, Oct. 2026.
2. J. Sun, K. Lukander, and B. U. Cowley, "Staged rule acquisition in WishGLD: Dissociation and compression across learning," in *9th International Conference on Motivational and Cognitive Control (MCC)*, Poster presentation, Helsinki, Finland, Aug. 2026.

Research Experience

Jun 2025 – Present

Research Assistant, HiPerCog Group, Department of Education, University of Helsinki (Supervisor: Ben Cowley).

- Designed and conducted an independent reanalysis of naturalistic fMRI data (Sherlock; Chen et al., 2017; N = 17) on how long-range narrative relations are represented. Built the full pipeline: pair- and time-resolved RSA, order-preserving permutation nulls, effect-injection power analysis and confound controls. Recalling a payoff selectively reinstated its earlier set-up in posterior medial cortex, whereas viewing showed no sustained relation signal. Poster, Brain & Mind Symposium 2026.
- Designed and implemented a modified WCST in PsyToolkit with an orthogonal orientation rule, enabling future reinforcement-learning modeling of multidimensional strategy use.
- Built the human-side analysis pipeline for a rat–human Go/NoGo attentional set-shifting paradigm: preprocessing from raw behavioral output, and a model-fitting pipeline matched to the existing rodent pipeline so that both species could be fitted under a single model.

Mar 2021 - Jun 2022

Undergraduate Research Assistant, Institute of Neuroscience, Chinese Academy of Sciences (Supervisor: Ye Zheng)

- Quantified Locus Coeruleus/Substantia Nigra integrity from neuromelanin-MRI via manual ROI tracing and intensity normalization. Linked LC structural metrics to set-shifting deficits in Parkinson's disease (CANTAB IED)
- Assisted in collecting and organizing neuropsychological data from DBS patients, contributing to analyses of stimulation-related cognitive impairment in GPi/STN cohorts. (Collaborator: Dr. Wu Jianjun)

Skills

Languages — Mandarin (Native), English [TOEFL iBT: 112/120 (R28, L30, W30, S24)]

Coding — Python, R, MATLAB

Analytical Skills — Python (pandas, NumPy, scipy, matplotlib);
EEG (MNE-Python, EEGLAB, ERP/time-frequency analysis, ISC, IS-RSA);
Statistics/modeling (GLM, PCA, HMM, mixed-effects models, permutation testing, FDR correction);
fMRI (GLM, MVPA/decoding, RSA - independently implemented)
NLP (Basic)

Academic Awards

2024-2026 — **100% Tuition Fee Waiver**, University of Helsinki, Master's Programme in Neuroscience

Research-related Activities

Oct 2025 - Jun 2026 **Curriculum Reviewer,** Master's Programme in Neuroscience, University of Helsinki

- Selected student reviewer for evaluating and redesigning the online course **Systems & Cognitive Neuroscience**.
- Provided pedagogical feedback on structure, clarity, and accessibility for students with diverse academic backgrounds.
- Contributed to improving learning alignment and conceptual scaffolding for future cohorts.

Jan 2025 - Jun 2026 **Student Representative,** Master's Programme in Neuroscience, University of Helsinki

- Served as student representative on the programme board; communicated student needs and coordinated curriculum improvements.
- Synthesized student feedback into actionable recommendations that informed course design, assessment methods, and program structure.